# DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT. – 2026/2027.

## Linear Algebra

### Algebra of matrices

Lecture No. 02



By- Dr. Puneet Sharma Sir

# Recap of previous lecture



**Topic**

ALGEBRA of MATRICES

# Topics to be Covered



**Topic**

TYPES of MATRICES

## Conjugate of Comp. Number →

$$z = x + iy$$

$$\Rightarrow \bar{z} = x - iy$$

eg $z_1 = 2 - 3i$ , $\bar{z}_1 = 2 + 3i$

$z_2 = 4i$ , $\bar{z}_2 = -4i$

$z_3 = -7 + 2i$, $\bar{z}_3 = -7 - 2i$

$z_4 = 3$ , $\bar{z}_4 = 3$

## Conjugate Mat & Transpose Mat →

$$A = \begin{bmatrix} 2 & 4-i & -3i \\ 4+i & 0 & -7 \end{bmatrix}_{2\times 3}$$ then $\bar{A} = ?$ & $A^T = ?$

$$\bar{A} = \begin{bmatrix} 2 & 4+i & +3i \\ 4-i & 0 & -7 \end{bmatrix}_{2\times 3}$$ & $$A^T = \begin{bmatrix} 2 & 4+i \\ 4-i & 0 \\ -3i & -7 \end{bmatrix}_{3\times 2}$$

## Transjugate Mat (Transposed Conj. Mat) →

$$A^\theta = \overline{(A^T)} \text{ or } (\bar{A})^T$$

$$A^\theta = \begin{bmatrix} 2 & 4-i \\ 4+i & 0 \\ 3i & -7 \end{bmatrix}$$

**Real Matrix**: If in a Mat, all the entries are Real Nos then Mat is Called Real Mat.

ie if $A$ is Real Mat then $\bar{A} = A$ (or $A^\theta = A^T$) Learn.

eg $A = \begin{bmatrix} 2 & -3 & 4 \\ 1 & 2 & 0 \end{bmatrix}$ = Real Mat (ie $A^\theta = (\bar{A})^T = A^T$)

eg $A = \begin{bmatrix} 2 & -3 & 4i \\ 1 & 2-i & 0 \end{bmatrix}$ = Complex Mat ie $\bar{A} \neq A$

**Complex Mat** → If in a Mat, at least one element is Complex number then Matrix is Called Complex Mat.

**Reversal Law** →

$$\left.\begin{aligned} (ABC)^T &= C^T B^T A^T \\ (ABC)^\theta &= C^\theta B^\theta A^\theta \\ (ABC)^{-1} &= C^{-1} B^{-1} A^{-1} \end{aligned}\right\}$$ & these Results Can also be generalised.

$A^{-1}, B^{-1}, C^{-1}, D^{-1}$ exist

**Q** If $A, B, C, D$ are Non singular Matrices of same order s.t

$ABCD = I$ then $B$ & $B^{-1} = ?$

**Sol:**

**(M-I)** $B = \frac{I}{ACD} = I \cdot A^{-1} C^{-1} D^{-1} = A^{-1} C^{-1} D^{-1}$

Blunder

**(M-II)**

$$A^{-1}(ABCD)D^{-1} = A^{-1} I D^{-1}$$

$$I \cdot BC \cdot I = A^{-1} D^{-1}$$

$$BC = A^{-1} D^{-1}$$

$$(BC)C^{-1} = A^{-1} D^{-1} C^{-1}$$

$$B = A^{-1} D^{-1} C^{-1}$$

**(ii)** $B = A^{-1} D^{-1} C^{-1}$

$$B^{-1} = (A^{-1} D^{-1} C^{-1})^{-1}$$

$$= (C^{-1})^{-1} (D^{-1})^{-1} (A^{-1})^{-1} \quad (\text{By R Law})$$

$$B^{-1} = C \cdot D \cdot A$$

Ans

# Some Special Matrices →

$\Rightarrow |A| = 0$

$$\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}, \quad \begin{bmatrix} 0 & h & -g \\ -h & 0 & f \\ g & -f & 0 \end{bmatrix}_{3\times3}, \quad \begin{bmatrix} 2 & 4+i & -5 \\ 4-i & -3 & 2-3i \\ -5 & 2+3i & 0 \end{bmatrix}, \quad \begin{bmatrix} 2i & 4+i & -5 \\ -4+i & -3i & 2-3i \\ 5 & -2-3i & 0 \end{bmatrix}$$

Symmetric ($\because A^T = A$)

Skew Symm. ($\because A^T = -A$)

HERMITIAN ($\because A^\theta = A$)

Skew Hermitian ($\because A^\theta = -A$)

Sp Note → If $A_{n\times n}$ & $A$ is Skew Symm

then $|A| = \begin{cases} 0, & n = \text{odd} \\ \text{Perfect Square}, & n = \text{even} \end{cases}$

"Det of Skew Symm Mat of ODD order is always Zero"

eg $A = \begin{bmatrix} 0 & 1 & 2 & 0 \\ -1 & 0 & 4 & 0 \\ -2 & -4 & 0 & 6 \\ 0 & 0 & -6 & 0 \end{bmatrix}_{4\times4}$ then $|A| = ?$

= skew symm. Mat

(a) 7 (c) -3

(b) 2 (d) None $\left[\because |A| = 36 = (6)^2\right]$

Note (1) Symmetric Mat is symmetrical about leading diagonal

(2) In skew symmetric Mat, Diagonal elements are all Zero & Corresponding elements are of opposite sign.

(3) In Hermitian Matrix, Diagonal elements are either Zero or purely Real Nos & Corresponding elements are Conjugates of each other.

(4) In skew Hermitian Mat, Diagonal elements are either Zero or purely Imaginary Nos & Corresponding elements are -ve Conjugates of each other.

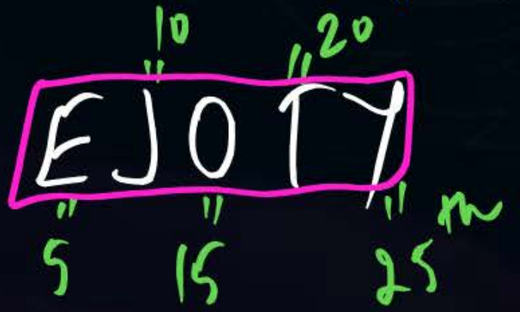
* Maximum Number of Different elements that will be Required to Construct Symmetric Mat of order $n \times n = \frac{n(n+1)}{2}$ (Learn)

eg $A_{2\times2} = \begin{bmatrix} a & c \\ c & b \end{bmatrix}$ ⇒ Max 3 elements are Required i.e $3 = 1+2$

eg $A_{3\times3} = \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & b \end{bmatrix}$ ⇒ Max 6 elements are Req i.e $6 = 1+2+3$

eg $A_{4\times4} = \begin{bmatrix} a & b & c & d \\ b & e & f & g \\ c & f & h & i \\ d & g & i & j \end{bmatrix}_{4\times4}$ ⇒ Max 10 elements are Req. & So on - - - -

i.e $10 = 1+2+3+4$



⊛ Every Square Mat $A_{n\times n}$ can be expressed as a sum of symmetric & skew symm Matrix. i.e

$$A = \left(\frac{A+A^T}{2}\right) + \left(\frac{A-A^T}{2}\right)$$

$$A = P + Q \quad \text{---(1)}$$

**Sq. Mat = Symm + Skew Symm**

$$P^T = \left(\frac{A+A^T}{2}\right)^T = \frac{A^T+(A^T)^T}{2} = \frac{A^T+A}{2} = P \text{ is P is symm.}$$

& $$Q^T = \left(\frac{A-A^T}{2}\right)^T = \frac{A^T-(A^T)^T}{2} = \frac{A^T-A}{2} = -\left(\frac{A-A^T}{2}\right) = -Q \text{ is Q is skew symm.}$$

(*) Every square Mat can be expressed as a sum of Hermitian & Skew Hermitian Matrix.

ie $$A = \left(\frac{A+A^\theta}{2}\right) + \left(\frac{A-A^\theta}{2}\right)$$

$$A = R + S$$

**Sq. Mat = Herm Mat + Skew Herm Mat**

$$R^\theta = \left(\frac{A+A^\theta}{2}\right)^\theta = \frac{A^\theta + (A^\theta)^\theta}{2} = \frac{A^\theta + A}{2} = R \Rightarrow R \text{ is Hermitian}$$

$$S^\theta = \left(\frac{A-A^\theta}{2}\right)^\theta = \frac{A^\theta - (A^\theta)^\theta}{2} = \frac{A^\theta - A}{2} = -\left(\frac{A-A^\theta}{2}\right) = -S \Rightarrow S \text{ is skew Hermitian}$$

Q. How many Different Symmetric Matrices of $n \times n$ are possible having entries either 0 or 1 or 2 ?

(a) $3^{n^2}$

(b) $\frac{n(n+1)}{2}$

(c) $2^{\frac{n(n+1)}{2}}$

(d) $3^{\frac{n(n+1)}{2}}$ ✓

Total Number of places which we have to fill $= \frac{n(n+1)}{2}$

Number of ways to fill each place $= 3$ ways

Total No. of ways to fill all places

$= \underbrace{3 \times 3 \times 3 \times 3 \times \ldots\ldots \times 3 \times 3}_{\frac{n(n+1)}{2} \text{ places}}$ ways $= 3^{\frac{n(n+1)}{2}}$

Q: If $A=[a_{ij}]_{n\times n}$ s.t $a_{ij}=i-j$ $\forall i$ & $j$ then $\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}=?$

(a) $n^2-n$

(b) $n^2$

(c) $\frac{n(n+1)}{2}$

(d) 0

Let $n=4$, $A=[a_{ij}]_{4\times4}=\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}=\begin{bmatrix} 0 & -1 & -2 & -3 \\ 1 & 0 & -1 & -2 \\ 2 & 1 & 0 & -1 \\ 3 & 2 & 1 & 0 \end{bmatrix}$ = Skew Symm. Matrix

$\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}=a_{11}+a_{12}+a_{13}+\ldots\ldots+a_{44}=0$

(d) 0 i.e Sum of all the elements of skew symm of any order = ZERO

(ii) In above Question if $n\geq3$, $n$=odd then $\vec{A}^{-1}=?$

Possible values of $n=3,5,7,9,\ldots\ldots$

Let $n=3$

$A=[a_{ij}]_{3\times3}$ = Skew Symm, $\because |A|=0\Rightarrow \vec{A}^{-1}$ = ONE

Q. if A & B are Hermitian Matrices of same order then $(AB-BA)=?$

(a) Symm. Mat

(b) Skew Symm Mat

(c) Hermitian Mat

(d) Skew Herm. Mat.

Given $A^\theta = A$ & $B^\theta = B$.

Now, $(AB-BA)^\theta = (AB)^\theta - (BA)^\theta$

$= B^\theta A^\theta - A^\theta B^\theta$ (By R. Law)

$= BA - AB$

$= -(AB-BA)$

$\Rightarrow (AB-BA)$ is skew Hermitian

Q: Extract Symm Mat, Anti Symm Mat, Herm Mat, Skew Herm Mat from the Mat

$$A=\begin{bmatrix} 2 & 4+2i & -6i \\ 0 & -4i & 2+4i \\ 6 & 8 & -4 \end{bmatrix}_{3\times 3}$$

Symm Mat $= \frac{A+A^T}{2} =$

Skew Symm $= \frac{A-A^T}{2} =$

HW

Herm. Mat $= \frac{A+A^\theta}{2} =$

Skew Herm Mat $= \frac{A-A^\theta}{2} =$

**Idempotent Mat** → if $A^2 = A$ ⟺ then A is called Idempotent

$\Rightarrow |A|^2 = |A| \Rightarrow |A|(|A|-1) = 0 \Rightarrow |A| = 0$ or $1$

**Involutary Mat** → if $A^2 = I$ ⟺ then A is called Involutary

$\Rightarrow |A|^2 = |I| \Rightarrow |A|^2 = 1 \Rightarrow |A| = 1$ or $-1$

**Nilpotent Mat** if $A^K = O$ (where k is least the integer)

⟺ then A is called N-Mat of index k.

eg if A is such that $A^2 \neq 0$, $A^3 \neq 0$, But $A^4 = 0$ then $A^5 = A^6 = A^7 = \ldots = O$

So A is N. Mat of index four

$(\because A^5 = A.A^4 = A.O = O)$

Note: **Identity** Matrix can be assumed as both Idempotent as well as Involutary.

Q: if A is an Idempotent Mat then $(I-A)$ will be ?

(a) Null Mat

(b) Involutary

(c) ✓ Idempotent

(d) Singular Mat

Given $A^2 = A$ — (1)

$A \cdot A = A$

$A = I$ ) WRONG step

Hence

$I - A = I - I = 0$

(M-II)

Given $A^2 = A$ — (1)

$(I-A)^2 = (I-A)\cdot(I-A)$

$= I^2 - IA - AI + A^2$

$= I - A - A + (A)$

$= (I-A)$

$\Rightarrow (I-A)$ is also Idempotent Mat

(M-III)

Given $A^2 = A$ — (1) Now $I - A = I - A^2 = ??$ Now what to Do??

**Orthogonal Mat →** (Mat A is Called orthogonal Mat) if ($AA^T = I$) (or $A^{-1} = A^T$)

OR

if $AA^T = I$ ($\text{or } A^{-1} = A^T$) $\Longleftrightarrow$ then A is Called orthogonal Mat.

**Unitary Mat →** if $AA^\theta = I$ ($\text{or } A^{-1} = A^\theta$) $\Longleftrightarrow$ then A is Called Unitary Mat.

**Note →** ① Det of an Orthogonal Mat is either 1 or −1

Let A is an orthogonal Mat then $AA^T = I$

$|AA^T| = |I|$

$|A||A^T| = 1$

$|A||A| = 1$

$|A|^2 = 1$

$|A| = \pm 1$

Hence Proved.

② orthogonal & Unitary Mat are always Invertible (True)

∵ For O-Mat, $A^{-1} = A^T$

& for U-Mat, $A^{-1} = A^\theta$

## (3) Unitary Mat of Real Nos is orthogonal also.

Let A is Unitary Mat then $AA^{\theta} = I$

Let A is also Formed by Real Nos

then A is Real Mat also $\Rightarrow A^{\theta} = A^{T}$

$\Rightarrow \boxed{AA^{T} = I}$

which is the Condition of Orthogonal Mat.

**Q** P.T $A = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$ is an Orthogonal Mat.

**Sol:** $\because AA^{T} = \begin{bmatrix} A \end{bmatrix}\begin{bmatrix} A^{T} \end{bmatrix} = \cdots\cdots = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$

Hence Proved

HW

Ms 8

If $A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ then A is ?

(a) Hermitian Mat $(\because A^{\theta} = A)$

(b) Involutary Mat $(\because A^2 = I)$

(c) Unitary Mat $(\because A A^{\theta} = I)$

(d) orthogonal Mat

# THANK - YOU

Tel: drpuneet Sir pw